

INDIAN STATISTICAL INSTITUTE
Bachelor of Statistical Data Science, Third Year
First Semester, 2026-27
Statistical Learning I
Mid-semester Examination

Time: 1 Hour

Part-A

Full Marks:15

1. Consider a regression problem involving a response variable Y and two explanatory variables X_1 and X_2 . Suppose that there are n observations satisfying $\sum_{i=1}^n X_{1i} = \sum_{i=1}^n X_{2i} = \sum_{i=1}^n X_{1i}X_{2i} = 0$. If the ordinary least squares method is used to fit the multiple linear regression model of Y on X_1 and X_2 , the fitted regression equation turns out to be

$$\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2.$$

A student, who is not familiar with multiple linear regression, attempts to estimate the regression coefficients using two successive simple linear regressions as follows.

First, the student regresses Y on X_1 and obtains the fitted regression equation

$$\hat{Y} = \alpha_0 + \alpha_1 X_1.$$

The residuals are then computed as $Y_i^* = Y_i - \alpha_0 - \alpha_1 X_{1i}$, for $i = 1, 2, \dots, n$. Next, the student regresses Y^* on X_2 and obtains the fitted regression equation

$$\hat{Y}_i^* = \gamma_0 + \gamma_2 X_{2i}.$$

Then the student concludes that the coefficients in the multiple regression model are given by

$$\beta_0 = \alpha_0 + \gamma_0, \quad \beta_1 = \alpha_1, \quad \beta_2 = \gamma_2.$$

Assume that a total of 6 marks are assigned to the question, with 2 marks each for correctly obtaining β_0 , β_1 , and β_2 . How many marks should the student receive? Justify your answer mathematically. [6]

2. Consider the data set consisting of the following eight observations on two variables X and Y .

X	-3	-2	-1	0	1	2	3	4
Y	-5	-3	0	1	3	5	7	8

Find (with proper mathematical justification) a value of θ that minimizes $\sum_{i=1}^8 |Y_i - \theta X_i|$. [4]

3. Consider a regression problem involving no intercepts. Assume that the response variable Y and the two covariates X_1 and X_2 , all are centred (i.e., the sample mean of each variable is 0). Also assume that the design matrix $\mathbf{X}$ satisfies $\mathbf{X}^\top \mathbf{X} = \mathbf{I}_2$, the 2×2 identity matrix.

If the ordinary least squares fit of Y on X_1 and X_2 is $\hat{Y} = 1.2X_1 - 0.5X_2$, find the values of β_1 and β_2 that minimize $\sum_{i=1}^n (Y_i - \beta_1 X_{1i} - \beta_2 X_{2i})^2$ subject to each of the following constraints:

(a) $|\beta_1| + |\beta_2| \leq 0.5$. [3]

(b) $0.5 \leq \beta_1^2 + \beta_2^2 \leq 1$ [2]

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(Answer as many as you can. The maximum you can score is 15.)

1. Consider a regression problem based on the following data set.

X	0.3	0.5	0.7	1.0	1.2	1.5	1.8	2.3	2.6	3.0
Y	3.6	3.3	3.1	2.9	3.3	3.6	a_1	a_2	4.1	3.6

Let $\hat{f}_h(\cdot)$ be the Nadaraya-Watson estimate of the regression function based on the Gaussian kernel with bandwidth h . If $\lim_{h \downarrow 0} \hat{f}_h(2) = 4$ and $\lim_{h \uparrow \infty} \hat{f}_h(2) = 3.6$, find (with proper justification) the values of a_1 and a_2 . [4]

2. Give suitable examples (with proper justification) of regression/classification problems, where

- (a) projection pursuit regression is expected to yield much better than additive regression.
- (b) distributions of the two competing classes are absolutely continuous, but the Bayes rule is not unique.
- (c) distributions of the two competing classes are absolutely continuous, no classifier can have misclassification probability smaller than $1/3$.

[3 × 3=9]

3. In order to construct a classification tree for a two-class classification problem, one needs to find the best split based on each of the measurement variables. Consider a categorical measurement variable C , which has three categories C_1 , C_2 , and C_3 . The numbers of observations in these three categories are given below:

Category	C_1	C_2	C_3	Total
No. of observations from Class 1	75	100	75	250
No. of observations from Class 2	0	175	75	250
Total	75	275	150	500

Among all possible binary splits based on C , determine the split that is optimal for every concave and symmetric impurity function ψ . Justify your answer. [5]